

Human-Computer Interaction: Chapter 8

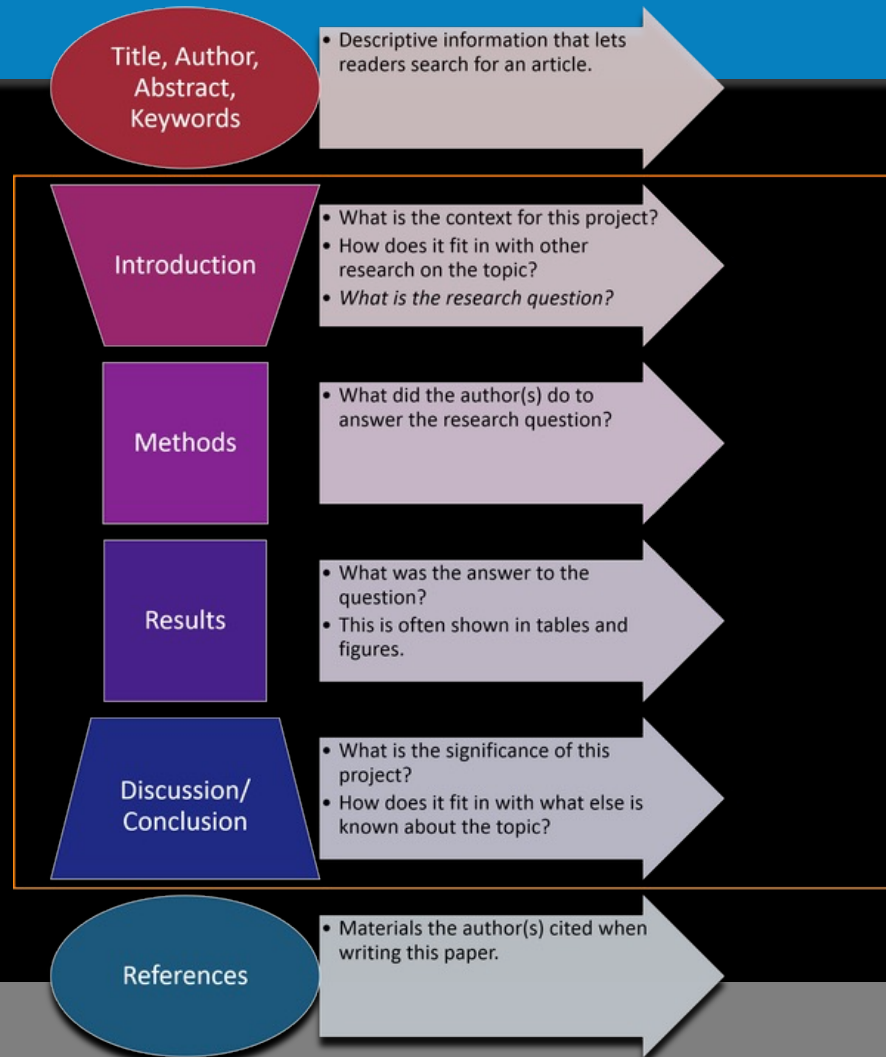
Writing
RESEARCH
PAPER



Conference vs. Journal?

- Both journals and conferences serve as platforms for dissemination of research findings.
- Depending on the domain, they differ significantly in their publication process (manuscript format, review timeline, etc.), evaluation criteria, and impact.
- Journal papers undergo a rigorous peer-review process, emphasizing in-depth analysis, while conference papers focus on novel developments and preliminary findings.
- In the HCI domain, some conferences such as SIGCHI are so prestigious and competitive that having a manuscript accepted there is deemed as important as publishing in a journal.

Parts of a research paper



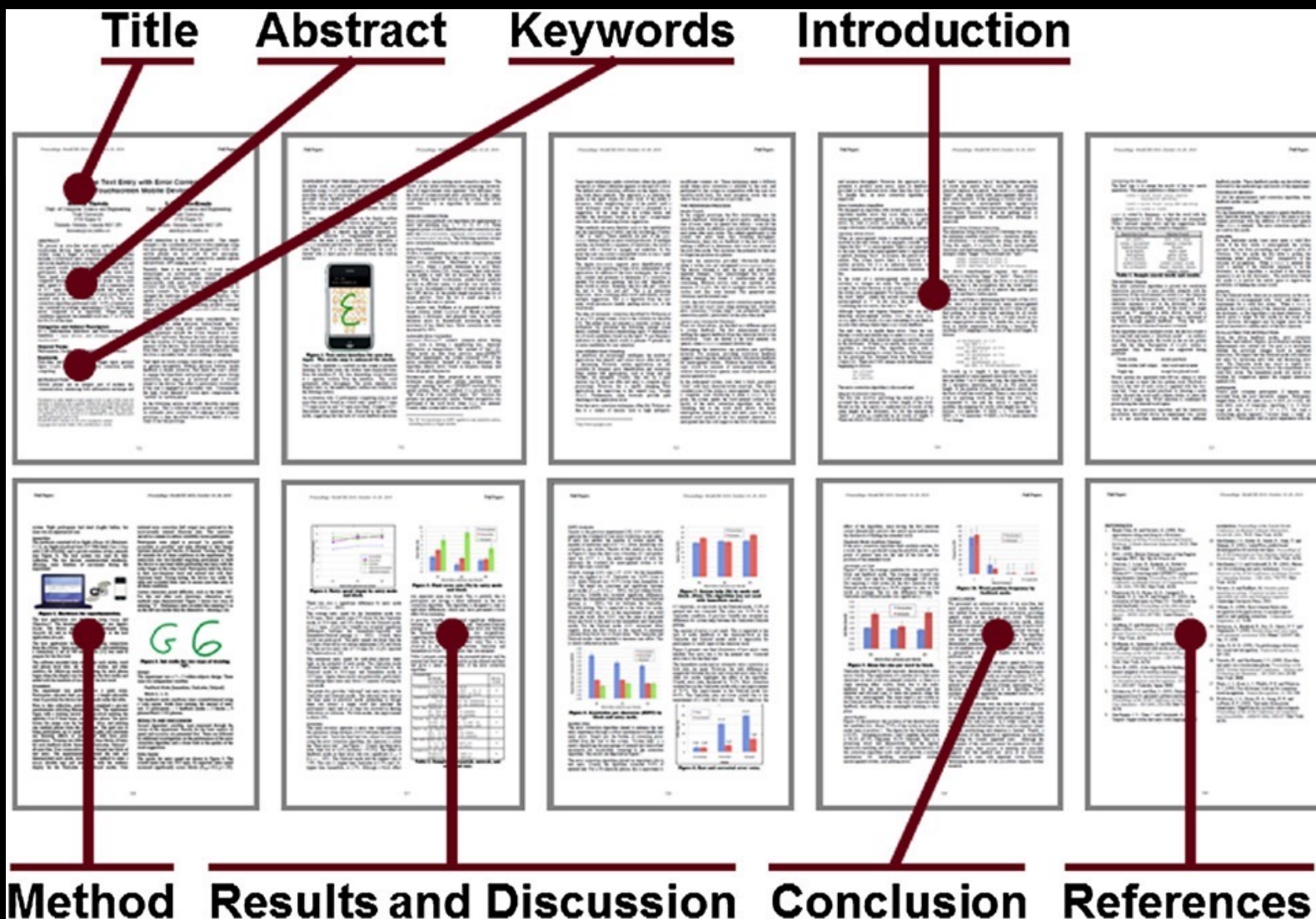


Figure 8.1. Parts of a research paper.

Parts of a research paper

Each part of the manuscript serves a purpose:

- **Title:** tells it all in one sentence.
 - Not too short and not too long.
 - Not too broad and not too specific.
 - People often try to make it catchy!
- **Abstract:** A one paragraph summary of the whole study that is often written last.
 - The abstract should tell the reader what was done and what was found.
 - Follows the same structure; intro, methods, results and implications.
 - For journal publications, the abstract is the most important element when it comes to the editor's first impression and decision.

Parts of a research paper

- **Keywords:** 3 to 6 most important terms that describe the theme of the research paper.
 - Is used for database indexing and searching
 - Example: Child-Robot Interaction, Mathematics learning, robot tutors, Motivational feedback, Facial expressions
- **Introduction:** The opening of the paper where you give the problem statement, motivation of research and the impact.
 - You can choose to present your literature review in the introduction or as a separate section e.g., “background” or “related works”.
 - The intro should be balanced in terms of breath and width of the reviewed topics.

Parts of a research paper

- **Method:** documents what you actually did in your study. Did you design a new interface? Did you test the interface in a user study? Whichever it is you should document all the steps you took so that your study is replicable.
- The methods section usually has multiple subheadings where you break down the information for the reader.
- How you structure the information is up to you for as long as the information flow is understandable for the reader.
- An HCI user study report often has the following subsections:
 - Participants
 - Experimental design
 - Materials and apparatus
 - Experimental procedure
 - Measurements
 - Analysis
 - etc.

Parts of a research paper

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- An HCI user study report often has the following subsections:

Remember! The language of the Methods (and Results section) is most often in past tense!

Example: We designed X... We conducted a study with ... We collected questionnaire data from... We analyzed the data using...

Parts of a research paper

- **Results:** Where you present the findings of your study and data analysis (e.g., data from questionnaires or interviews + statistical analysis)
 - It is common practice to use subsections that organize the results by dependent measures (Order them consistently with the order you defined them in the Methods section).
 - You must use visual aids (figures, tables, etc.) to make the results more engaging and understandable for the reader.
 - The Results and Discussion sections are at the same level as Method (So, You are not allowed to introduce results in the Methods section).
 - Also, you are not allowed to introduce new methods information in the results section.

Parts of a research paper

- **Discussion:** Tries to interpret the results reported in the Results section and make sense of them (What caused the differences in the measurements across experimental conditions? If no differences were observed, what could have been the reason? What were the limitations of the study, interface design or measurements? What could be improved in the future?)
 - The discussion section should link the findings of the study with previous literature supported by references. (was the interaction design better/more effective/etc.? Was the RQ answered? Was the gap in the lit. filled?)
 - Some time Discussion and Conclusion are presented together in some papers. It's generally advised to have a separate conclusion section at the end.
 - You are not allowed to introduce new results in the Discussion section

Parts of a research paper

- **Conclusion:** summarizes what the study did and restates the important findings.
 - Conclusion is like an abstract with more focus placed on the contribution of the paper to the field and suggestions for future.
 - Like abstract it shouldn't exceed one paragraph.
- References: a list of papers that were used and cited in the paper.
 - Only papers cited in the paper are included in the reference list.
 - If a paper is included in the references, it should be cited within text.
 - References and citations should be formatted as stipulated by the conference or journal.
e.g., References can be numbered or ordered alphabetically according to the first author's surname.

Preparing your manuscript

- Be mindful of formatting:
There is usually a template that you should follow!
- Check the book on referencing, citation and quotation guidelines
- Write clearly! Avoid long and complex sentence! Avoid passive voice as much as possible. Define abbreviations in their first appearance.
- Prepare sufficient visual aids
- Try to get ideas down first and then worry about the rest!

